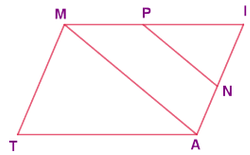
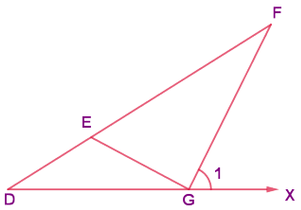
Challenging problems

6. In the figure, $\angle TMA \equiv \angle IAM$ and $\angle TAM \equiv \angle IMA$. P is the midpoint of MI and N is the midpoint of AI. Prove that $\Delta\ PIN \sim \Delta\ ATM$.

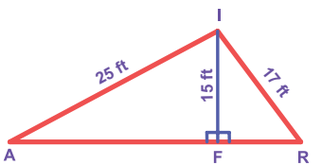


7. In the figure, if $\angle FEG \equiv \angle 1$ then, prove that $DG^2 = DE.DF$.



8. The diagonals of the rhombus is 12 *cm* and 16 *cm*. Find its perimeter. (Hint: the diagonals of rhombus bisect each other at right angles).

9. In the figure, find AR.



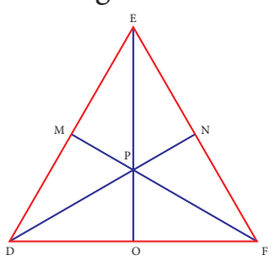
10. In ΔDEF, DN, EO, FM are medians and point P is the centroid. Find the following.

(i) IF DE = 44, then DM = ?

(ii) IF PD = 12, then PN = ?

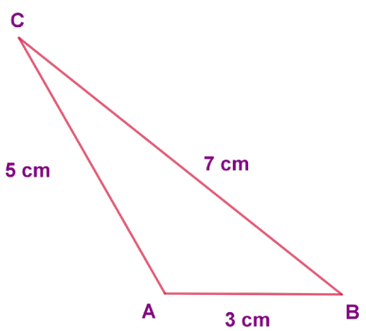
(iii) If DO = 8, then FD = ?

(iv) IF OE = 36 then EP = ?

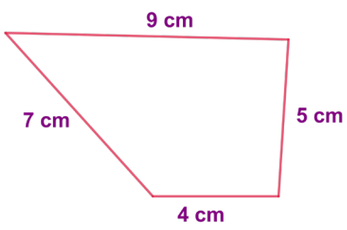


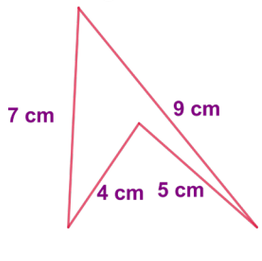
5.10 Construction of Quadrilaterals

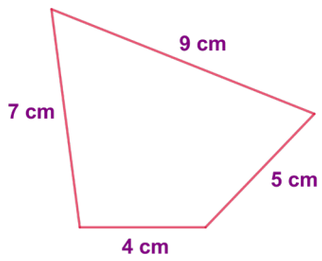
We have already learnt how to draw triangles in the earlier classes. A polygon that has got 3 sides is a triangle. To draw a triangle, we need 3 independent measures. Also, there is only one way to construct a triangle, given its 3 sides. For example, to construct a triangle with sides 3*cm*, 5*cm* and 7*cm*, there is only one way to do it.



Now, let us move on to quadrilaterals. A polygon that is formed by 4 sides is called a quadrilateral. Isn't it? But, a quadrilateral can be of different shapes. They need not look like the same for the given 4 measures. For example, some of the quadrilaterals having their sides as 4 *cm*, 5 *cm*, 7 *cm* and 9 *cm* are given below.







So, to construct a particular quadrilateral, we need a 5th measure. That can be its diagonal or an angle measure. Moreover, even if 2 or 3 sides are given, using the measures of the diagonals and angles, we can construct quadrilaterals.

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